

AI Investment Research Copilot - Project Roadmap

Part 1: Foundation & Infrastructure (Week 1)

- Backend: FastAPI, PostgreSQL, MongoDB, JWT Authentication, REST APIs.
- Frontend: React/Next.js, Tailwind CSS, authentication pages, dashboard.
- AI Infrastructure: LangChain, LangGraph, Vector DB, embeddings, RAG.
- DevOps: Docker, logging, environment configuration.

Part 2: Company Intelligence (Weeks 2–3)

- Company search by name, symbol, sector and industry.
- AI-generated company profile with business model, financials, CapEx, growth drivers, risks, competitors, bull/bear case and AI summary.
- Data ingestion from annual reports, filings, presentations, conference calls and news.

Part 3: Document Intelligence (Week 3)

- Upload PDFs, presentations and research reports.
- RAG pipeline with OCR, chunking, embeddings and Q&A.;

Part 4: Event Intelligence Engine (Week 4)

- Natural language event analysis.
- Output: event summary, beneficiary industries, companies, losers, risks, confidence score and reasoning.

Part 5: AI Investment Assistant (Week 5)

- Conversational AI for company, event and report analysis.

Part 6: News Intelligence (Week 6)

- Aggregate financial news, summarize, detect sentiment, map companies and sectors.

Part 7: Financial Analysis Engine (Week 7)

- Calculate key financial metrics and explain changes using AI.

Part 8: Valuation Engine (Week 8)

- DCF, Comparable Company Analysis, EV/EBITDA, PE, PB, PEG with AI explanations.

Part 9: Dashboard & Portfolio (Week 9)

- Market dashboard, watchlist, portfolio allocation and risk score.

Part 10: Alerts & Notifications (Week 10)

- Email, push and in-app alerts for news, earnings and policy updates.

Part 11: Authentication & User Features

- Email/Google login, profile, bookmarks, watchlist and search history.

Part 12: Deployment & Production

- Deploy backend, frontend, databases, AI services and monitoring.

Future Enhancements

- AI stock screener, insider trading tracker, technical analysis, scenario analysis, multilingual support, mobile apps and report generation.

Recommended Development Order

- Foundation → Company Intelligence → Document Intelligence → Event Intelligence → AI Assistant → News → Financial Analysis → Valuation → Dashboard → Alerts → Deployment.